ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION — DECEMBER / JUNE 2023

B.TECH. CSE ENGINEERING — SEMESTER 3

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted.

--- UNIT I ---

Q.1 Explain Propositional Logic, Truth Tables, Tautology, and Predicate Calculus with Quantifiers. [14 Marks]

[OR]

Q.1 (Alt) State Principle of Mathematical Induction and prove summation formulas. [14 Marks]

--- UNIT II ---

Q.2 Explain Relations: Equivalence relations, Equivalence classes, and Partial Order Relations (Posets, Hasse Diagrams). [14 Marks]

[OR]

Q.2 (Alt) Explain Functions: Injective, Surjective, Bijective. State Pigeonhole Principle and applications. [14 Marks]

--- UNIT III ---

Q.3 Define Groups, Subgroups, Cyclic Groups, and Lagrange's Theorem on cosets. [14 Marks]

[OR]

Q.3 (Alt) Explain Rings, Integral Domains, and Fields with algebraic property proofs. [14 Marks]

--- UNIT IV ---

Q.4 Solve Linear Recurrence Relations with constant coefficients using Characteristic equation and Generating functions. [14 Marks]

[OR]

Q.4 (Alt) State Inclusion-Exclusion Principle and Derangement formulas with counting problems. [14 Marks]

--- UNIT V ---

Q.5 Explain Graph Theory: Euler graphs, Hamiltonian paths, Planar graphs, and Kuratowski's theorem. [14 Marks]

[OR]

Q.5 (Alt) Describe Tree properties, Spanning trees, and Graph Coloring with Chromatic Number. [14 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****